

PETERBOROUGH TRENT UNIVERSITY, ON, Canada

WMO: 716720

Lat: 44.3511N

Lon: 78.2950W

Elev: 712

StdP: 14.32

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.7	(c) -3.7	(d) -17.2	(e) 2.2	(f) -8.4	(g) -11.6	(h) 3.0	(i) -2.0	(j) 16.4	(k) 24.6	(l) 14.2	(m) 24.2	(n) 2.8	(o) 290	(p) 0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	87.4	70.4	84.3	69.1	81.5	67.4	73.1	82.7	71.5	80.4	69.9	78.0	6.9	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.2	114.0	77.4	68.6	107.9	76.1	67.0	102.1	74.5	37.2	83.0	35.8	80.4	34.4	78.2	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.6	11.8	10.3	-19.2	92.1	3.9	3.4	-22.0	94.5	-24.3	96.5	-26.4	98.4	-29.3	100.9
			-19.5	76.2	3.8	2.1	-22.3	77.7	-24.5	78.9	-26.7	80.1	-29.5	81.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.6	18.8	19.8	29.5	42.7	55.7	63.7	69.0	66.8	59.6	47.7	35.6	25.0		
	DBStd	19.78	12.36	11.18	10.99	8.27	7.99	6.18	5.41	5.18	7.24	7.77	8.72	10.32		
	HDD50	4089	966	846	640	246	34	1	0	0	7	140	433	776		
	HDD65	7784	1431	1266	1101	669	305	95	22	40	195	538	881	1241		
	CDD50	2127	0	0	4	26	210	412	588	522	293	69	3	0		
	CDD65	347	0	0	0	0	17	56	145	96	32	1	0	0		
	CDH74	3595	0	0	2	11	262	653	1420	891	337	19	0	0		
	CDH80	929	0	0	0	0	58	165	414	217	74	1	0	0		
(14) Wind	WSAvg	4.6	4.9	5.3	5.2	5.5	4.6	4.1	3.9	3.7	3.7	4.5	4.6	4.7		
(15) Precipitation	PrecAvg	33.5	2.8	2.0	2.0	2.7	3.3	3.1	3.3	2.6	3.0	2.8	2.9	2.7		
	PrecMax	45.2	5.4	3.8	3.8	5.0	5.2	7.3	11.5	5.1	7.2	6.9	5.7	4.9		
	PrecMin	26.3	1.7	0.8	0.6	0.8	0.9	0.9	1.0	0.9	1.3	0.9	1.6	0.4		
	PrecStd	5.0	0.9	0.8	0.9	1.2	1.1	1.8	2.1	1.1	1.3	1.5	1.1	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.2	50.3	68.2	75.3	85.9	88.5	91.2	90.5	87.0	77.1	62.7	52.2		
		MCWB	46.5	46.1	54.0	58.5	69.8	70.0	72.4	70.6	71.2	65.4	52.8	49.9		
	2%	DB	43.4	43.7	55.6	69.3	80.9	84.6	88.0	85.3	81.7	70.2	57.1	46.2		
		MCWB	40.9	40.1	47.1	53.5	66.0	69.3	71.0	69.0	68.6	60.7	51.8	44.0		
	5%	DB	39.3	39.3	49.8	63.5	76.3	81.0	84.7	81.8	77.9	65.3	53.2	42.1		
		MCWB	37.1	36.2	43.0	49.8	61.9	67.0	69.2	67.3	66.2	57.6	48.2	39.7		
	10%	DB	35.8	35.9	44.7	58.4	71.9	77.4	81.5	79.1	73.8	60.6	49.3	38.3		
		MCWB	33.8	33.3	39.3	47.7	59.4	64.4	67.4	65.6	64.1	54.8	45.4	36.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.3	45.7	58.1	60.9	71.2	74.1	76.0	74.5	73.6	68.3	56.0	50.2		
		MCDB	48.0	47.9	66.6	71.5	82.9	83.0	86.9	83.8	82.8	74.2	58.8	51.8		
	2%	WB	41.6	40.5	49.1	56.4	67.8	71.6	73.4	72.1	71.1	62.6	52.5	44.3		
		MCDB	42.9	43.2	53.0	65.0	78.2	81.8	83.4	80.6	79.2	66.9	56.7	45.8		
	5%	WB	37.5	36.4	43.8	52.7	64.8	69.4	71.5	70.3	68.5	58.9	49.0	40.2		
		MCDB	39.3	38.7	48.5	60.9	73.2	77.5	80.1	77.7	74.7	64.0	52.3	42.1		
	10%	WB	33.9	33.5	39.4	48.9	61.5	67.0	69.8	68.4	65.7	55.5	45.7	36.3		
		MCDB	35.5	35.7	44.4	56.8	69.0	73.9	78.3	75.5	71.6	59.4	49.3	38.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	18.0	18.8	20.6	23.1	21.5	22.1	21.7	22.2	17.8	16.4	13.6		
		MCDBR	16.6	19.9	25.7	31.4	29.3	26.8	25.5	25.5	26.3	24.8	23.2	16.4		
	5% WB	MCWBR	15.3	17.2	17.4	17.8	15.0	13.1	11.4	11.6	13.9	16.1	17.8	14.7		
		MCWBR	16.7	18.9	23.6	26.1	25.3	22.7	21.6	20.2	22.7	22.6	21.0	15.7		
(40) Clear-Sky Solar Irradiance	taub		0.300	0.302	0.316	0.352	0.373	0.409	0.409	0.399	0.362	0.349	0.307	0.280		
	taud		2.322	2.323	2.423	2.390	2.373	2.289	2.300	2.321	2.420	2.448	2.513	2.449		
	Ebn at Noon		258	281	293	289	285	273	272	271	273	261	258	258		
	Edh at Noon		25	30	32	36	37	41	40	38	32	27	21	21		
(44) All-Sky Solar Radiation	RadAvg		429	711	1128	1411	1706	1833	1873	1633	1305	764	473	341		
	RadStd		45	67	104	151	180	149	139	82	115	71	59	42		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page